

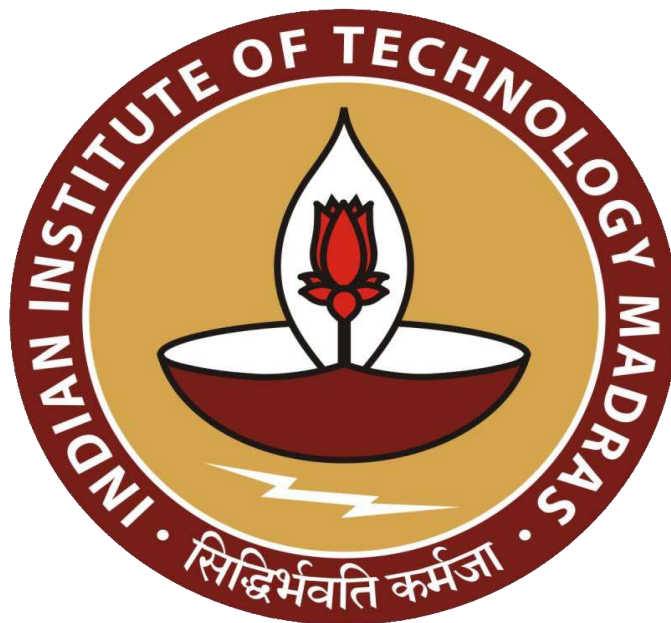
Data-Driven Optimization in Bottle Manufacturing

A Proposal report for the BDM capstone Project

Submitted by

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Contents

1	Executive Summary	3
2	Organization Background	3
3	Problem Statement	4
3.1	Problem statement 1:	4
3.2	Problem statement 2:	4
3.3	Problem statement 2:	4
4	Background of the Problem	4
5	Problem Solving Approach	5
6	Expected Timeline	6
7	Expected Outcome	7

Declaration Statement

I am working on a Project Title “**Data-Driven Optimization in Bottle Manufacturing**”. I extend my appreciation to **S P Enterprises** for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

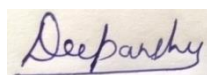
Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of Candidate:



Name: DEEPANSHU KAMBOJ

Date: 11/06/2025

1 Executive Summary:

S P Enterprises is a small manufacturing company that makes plastic bottles of various sizes from 100ml to 5L. They supply bottles to pharmaceutical and healthcare companies. The company is currently struggling with issues in raw material procurement, uneven fulfillment of client orders, and space constraints, all of which are affecting profitability and productivity.

The project aims to solve these problems by using business data analysis and insights from factory operations. By studying raw material purchase records and daily order data, we will look for patterns, forecast future needs, and understand how materials are used. We will clean the data and create visuals like graphs and charts to make the information easier to understand. The goal is to help SP Enterprises plan raw material purchases better, improve how and when they deliver orders to clients, utilize space in better way, and increase overall profits.

2 Organization Background :

S P Enterprises is located in Bhagwanpur, District Haridwar, Uttarakhand. The company was established three years ago on a 2,000 sq. ft. plot and operates as a manufacturing unit producing plastic bottles for pharmaceutical and healthcare packaging. Currently three skilled labour handles manufacturing process. S P Enterprises operates in the B2B segment, supplying bottles of various capacities like 100ml, 250ml, 500ml, 1L, 2L, and 5L to clients such as Ayurgram Health Care, Farmatech Pvt. Ltd., and Rasko India Vetcare. These bottles are primarily used for packaging liquids, syrups and medical solutions.

The company buys raw materials such as HDPE, PP, PET, HIPS, and PVC sheets from well known petrochemical suppliers like Haldia Petrochemicals, Reliance Industries, and Nobel Polymers. It follows a weekly schedule for buying raw material and making bottles depending on how many and how urgent client orders are. Production capacity of company is approximately 2000 bottles per day. The company is continuously working to improve how it works, reduce costs and deliver orders to clients on time.

3 Problem Statement:

3.1 Inefficient Raw Material Procurement Strategy: Raw materials are purchased weekly without accurate demand forecasting, which leads to overstocking or under stocking.

3.2 Demand-Supply Imbalance and Production Bottlenecks: Orders from clients are not evenly distributed, leading to overloaded production on some days and underutilized capacity on others.

3.3 Space Constraints: Due to limited space, storing plastic bottles within factory before shipment becomes difficult.

4 Problem Background:

Inefficient Raw Material Procurement Strategy:

The company buys raw materials every week, but there is no proper system to predict how much is actually needed. Sometimes they buy too much, which locks up money in extra stock. Other times, they buy too little, which delays production. These problems happen mainly because the planning is done manually and not based on proper data analysis. Also sometimes purchasing raw material from one supplier costs more as compare to other supplier.

Demand-Supply Imbalance and Production Bottlenecks:

The factory struggles to match supply with demand because client orders keep changing in size and timing. Since the factory can only make a fixed number of bottles each day, so sudden increase in orders puts extra pressure on machines and workers. On the other hand when there are few orders, the machines and workers stay idle, which wastes money. Also, the number of skilled workers is limited, so it is hard to manage when orders suddenly go up.

Space Constraints:

The growing volume of bottles leads to overcrowding, making movement and operations less efficient. This space limitation disrupts workflow, delays processing, and affects overall productivity. Without proper storage solutions, the workflow is disrupted and overall productivity is negatively impacted.

5 Problem Solving Approach

Data Collection:

The data required for this project includes weekly raw material purchase records and bottle order data for a 6-month period (may change based on availability). The purchase data consists of material types (HDPE, PET, PP, etc.), quantities, cost per kg, and supplier details which will be analyzed to compare price from different suppliers and assess whether quantity purchased aligns with order requirements. The order data includes bottle sizes (100ml to 5L), number of bottles demanded, number of bottles produced and client names which will be analyzed to see future demands using demand forecasting.

Methods of Analysis:

Raw Material Usage Analysis: Weekly raw material data for items like HDPE, PET, PP, PVC, and HIPS will be studied to understand how much is used and how prices change over time. The costs from different suppliers will be compared to find out who offers the best prices. This will help in checking if materials are bought at the right time and from right supplier. Simple charts like line graphs and moving averages will be used to show trends in material use and spot any sudden changes that might point to poor planning or sudden demand. Also, by understanding actual usage patterns, the company can avoid storing excess material, so this will help in saving warehouse space and improving material arrangement in storage areas.

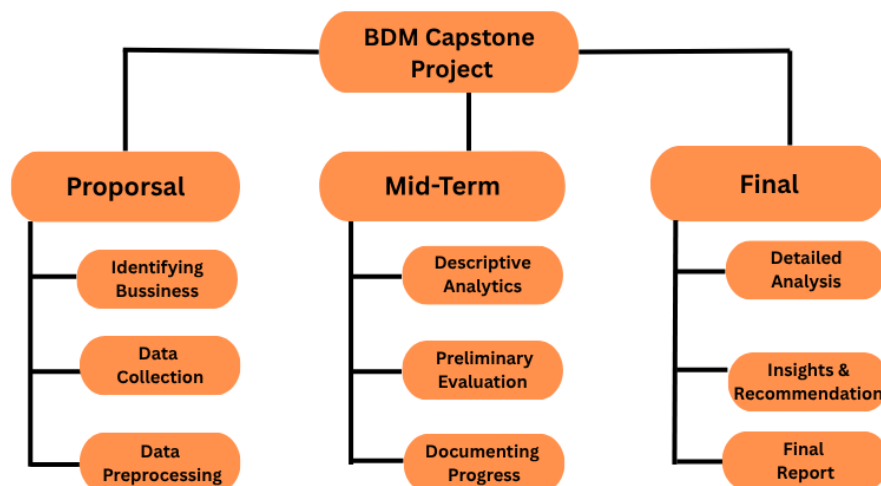
Demand Forecasting: The past order data will be used to guess future demand for bottles based on size and client. Simple methods like line trends or averages will help in finding patterns such as increasing orders or seasonal demand. This will help factory plan better, buy the right amount of materials on time and avoid delays or extra costs. With accurate demand predictions, the factory can avoid unnecessary storing of raw materials or finished products which can lead to better space usage and fewer storage issues.

Visualization:

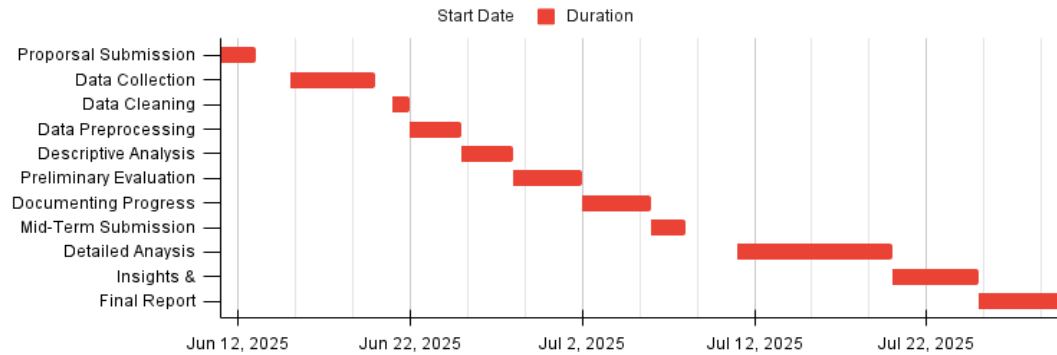
Key charts such as line graphs (for material trends), bar charts (for client comparison), and pie charts (for bottle size distribution) will be created. Additionally, pivot tables will help summarize weekly trends. Also scatter plot for seeing if there is any correlation with production variable prices. The final output will help the business in making informed decisions about procurement planning, production scheduling, and client targeting.

6 Expected Timeline

6.1 Work Breakdown Structure:



6.2 Gantt Chart:



7 Expected Outcome

- 7.1 Company will be able to reduce overstocking and under stocking issues.
- 7.2 Cash flow will be improved by purchasing only the required quantity.
- 7.3 Overall production efficiency will be improved.
- 7.4 Timely shipment and reduced delay in dispatch.
- 7.5 Company will be able to use available space in better way.